To find the extrema, we apply second derivative test. So,

$$\frac{d^2D}{dx^2} = 2\frac{y^2+x^2}{y^3}.$$

The value of $\left(\frac{d^2D}{dx^2}\right)_{\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)} > 0$; $\left(\frac{d^2D}{dx^2}\right)_{\left(-\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)} < 0$.

This implies the nearest and farthest points are $\left(\frac{1}{\sqrt{2}},\frac{1}{\sqrt{2}}\right)$ and $\left(-\frac{1}{\sqrt{2}},-\frac{1}{\sqrt{2}}\right)$ respectively.

Therefore, the nearest and the farthest distances are respectively $\sqrt{2}-1$ and $\sqrt{2}+1$.

Example 7.64

A steel plant is capable of producing x tonnes per day of a low-grade steel and y tonnes per day of a high-grade steel, where $y = \frac{40-5x}{10-x}$. If the fixed market price of low-grade steel is half that of high-grade steel, then what should be optimal productions in low-grade steel and high-grade steel in order to have maximum receipts.

Solution

Let the price of low-grade steel be ₹p per tonne. Then the price of high-grade steel is ₹$2p$ per tonne.

The total receipt per day is given by $R = px + 2py = px + 2p\left(\frac{40-5x}{10-x}\right)$. Hence the problem is to maximise R. Now, simplifying and differentiating R with respect to x, we get

$$R = p\left(\frac{80-x^2}{10-x}\right)$$

$$\frac{dR}{dx} = p\left(\frac{x^2-20x+80}{(10-x)^2}\right)$$

$$\frac{d^2R}{dx^2} = -\frac{40p}{(10-x)^3}$$

Now, $\frac{dR}{dx} = 0 \Rightarrow x^2-20x+80=0$ and hence $x = 10\pm 2\sqrt{5}$

At $x = 10-2\sqrt{5}$, $\frac{d^2R}{dx^2} < 0$

and hence R will be maximum. If $x = 10-2\sqrt{5}$ then $y = 5-\sqrt{5}$. Therefore the steel plant must produce low-grade and high-grade steels respectively in tonnes per day are

$$10-2\sqrt{5} \text{ and } 5-\sqrt{5}.$$

Example 7.65

Prove that among all the rectangles of the given area square has the least perimeter.

Solution

Let x, y be the sides of the rectangle. Hence the area of the rectangle is $xy = k$ (given). The perimeter of the rectangle P is $2(x+y)$. So the problem is to minimize $2(x+y)$ suject to the condition $xy = k$. Let $P(x) = 2\left(x+\frac{k}{x}\right)$.

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